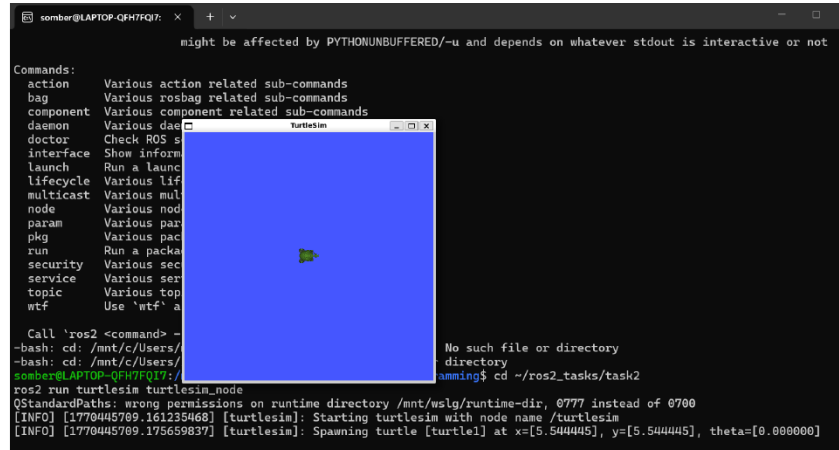


Task 2 - ROS2 (turtlesim) | Screenshot Set 1

```
git push -u origin main
On branch main
Your branch is up to date with 'origin/main'.

Untracked files:
  (use "git add <file>..." to include in what will be committed)
    lec2/

nothing added to commit but untracked files present (use "git add" to track)
[main c5f2fec] Add lec2 folder
1 file changed, 4 insertions(+)
create mode 100755 lec2/test_bash.sh
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 16 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (4/4), 369 bytes | 184.00 KiB/s, done.
Total 4 (delta 0), reused 0 (delta 0), pack-reused 0
To https://github.com/somber-xhp/S26_RobotProgramming_Iron.git
2160b32..c5f2fec main -> main
Branch 'main' set up to track remote branch 'main' from 'origin'.
somber@LAPTOP-QFH7FQ17:~/repos/S26_RobotProgramming_Iron$ cd ~/repos/S26_RobotProgramming_Iron
git remote -v
git log --oneline -n 3
origin https://github.com/somber-xhp/S26_RobotProgramming_Iron.git (fetch)
origin https://github.com/somber-xhp/S26_RobotProgramming_Iron.git (push)
c5f2fec (HEAD -> main, origin/main, origin/HEAD) Add lec2 folder
2160b32 first commit
```



The screenshot shows a terminal window with the following content:

```
might be affected by PYTHONUNBUFFERED/-u and depends on whatever stdout is interactive or not

Commands:
action      Various action related sub-commands
bag         Various rosbag related sub-commands
component   Various component related sub-commands
daemon      Various dae...
doctor      Check ROS s...
interface   Show inform...
launch      Run a launc...
lifecycle   Various lif...
multicast   Various mul...
node        Various nod...
param       Various par...
pkg         Various pac...
run         Run a packa...
security    Various sec...
service     Various ser...
topic       Various top...
ntf         Use 'ntf' a...

Call 'ros2 <command>' -
-bash: cd: /mnt/c/Users/somber/OneDrive/Desktop/ROS2_Tasks/task2: No such file or directory
-bash: cd: /mnt/c/Users/somber/OneDrive/Desktop/ROS2_Tasks/task2: No such file or directory
somber@LAPTOP-QFH7FQ17:~/repos/S26_RobotProgramming_Iron$ cd ~/ros2_tasks/task2
ros2 run turtlesim turtlesim_node
[StandardPaths: wrong permissions on runtime directory /mnt/wslg/runtime-dir, 0777 instead of 0700]
[INFO] [1770445709.161235468] [turtlesim]: Starting turtlesim with node name /turtlesim
[INFO] [1770445709.175659837] [turtlesim]: Spawning turtle [turtle1] at x=[5.544445], y=[5.544445], theta=[0.000000]
```

The turtlesim window is visible in the background, showing a blue square with a small green turtle icon in the center.

Related Questions & Answers

Q2. What are the executable nodes in the turtlesim package?

A: draw_square, mimic, turtle_teleop_key, turtlesim_node.

Q4. What is the name and starting pose of the simulated turtle?

A: Name: turtle1. Start pose: x=5.544445, y=5.544445, theta=0.000000 rad.

Task 2 - ROS2 (turtlesim) | Screenshot Set 2

```
somer@LAPTOP-QFH7FQI7: /mnt/c/Users/somer/Dropbox/RobotProgram$ cd ~/ros2_tasks/task2
ros2 run turtlesim turtlesim_node
QStandardPaths: wrong permissions on runtime directory /mnt/mslg
/runtime-dir, 0777 instead of 0700
[INFO] [1770445709.161235468] [turtlesim]: Starting turtlesim with node name /turtlesim
[INFO] [1770445709.175659837] [turtlesim]: Spawning turtle [turtle1] at x=[5.544445], y=[5.544445], theta=[0.000000]
[INFO] [1770445854.866319789] [turtlesim]: Rotation goal completed successfully
[INFO] [1770445857.809830586] [turtlesim]: Rotation goal completed successfully
[WARN] [1770445910.434061289] [turtlesim]: Rotation goal received before a previous goal finished. Aborting previous goal
[INFO] [1770445911.345467233] [turtlesim]: Rotation goal completed successfully
[INFO] [1770445916.978100960] [turtlesim]: Rotation goal completed successfully
[INFO] [1770445920.657073110] [turtlesim]: Rotation goal completed successfully
[INFO] [1770445921.841474743] [turtlesim]: Rotation goal completed successfully
[INFO] [1770445922.930093784] [turtlesim]: Rotation goal completed successfully

somer@LAPTOP-QFH7FQI7: /mnt/c/Users/somer/Dropbox/RobotProgramming$ cd ~/ros2_tasks/task2
ros2 node list
/turtle1
/teleop_turtle
/turtlesim

somer@LAPTOP-QFH7FQI7: ~/ros2_tasks/task2$ ros2 node info /turtlesim
/turtlesim
Subscribers:
  /parameter_events: rcl_interfaces/msg/ParameterEvent
  /turtle1/cmd_vel: geometry_msgs/msg/Twist
Publishers:
  /parameter_events: rcl_interfaces/msg/ParameterEvent
  /rosout: rcl_interfaces/msg/Log
  /turtle1/color_sensor: turtlesim/msg/Color
  /turtle1/pose: turtlesim/msg/Pose
Service Servers:
  /clear: std_srvs/srv/Empty
  /kill: turtlesim/srv/Kill
  /reset: std_srvs/srv/Empty
  /spawn: turtlesim/srv/Spawn
  /turtle1/set_pen: turtlesim/srv/SetPen
  /turtle1/teleport_absolute: turtlesim/srv/TeleportAbsolute
  /turtle1/teleport_relative: turtlesim/srv/TeleportRelative
  /turtlesim/describe_parameters: rcl_interfaces/srv/DescribeParameters
  /turtlesim/get_parameter_types: rcl_interfaces/srv/GetParameterTypes
  /turtlesim/get_parameters: rcl_interfaces/srv/GetParameters
  /turtlesim/get_type_description: type_description_interfaces/srv/GetTypeDescription
  /turtlesim/list_parameters: rcl_interfaces/srv/ListParameters
  /turtlesim/set_parameters: rcl_interfaces/srv/SetParameters
  /turtlesim/set_parameters_atomically: rcl_interfaces/srv/SetParametersAtomically
Service Clients:
Action Servers:
  /turtle1/rotate_absolute: turtlesim/action/RotateAbsolute
```

Related Questions & Answers

Q9. What happens if you try to drive the turtle out of the window?

A: The turtle cannot leave the window; it hits the boundary and will not move beyond the visible area.

Q2. How many nodes are running?

A: 2 nodes: /turtlesim and /teleop_turtle.

Q3. For moving and tracking, what does /turtlesim send and receive? Topic names?

A: Receives velocity commands on /turtle1/cmd_vel (geometry_msgs/msg/Twist); publishes pose on /turtle1/pose (turtlesim/msg/Pose).

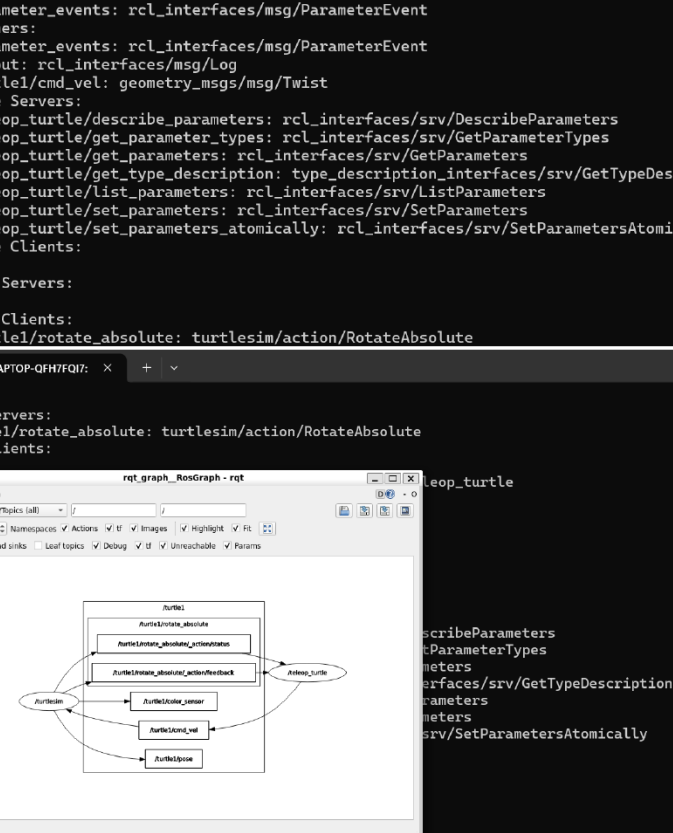
Task 2 - ROS2 (turtlesim) | Screenshot Set 3

```
somber@LAPTOP-QFH7FQI7:~/ros2_tasks/task2$ ros2 node info /teleop_turtle
/teleop_turtle
Subscribers:
  /parameter_events: rcl_interfaces/msg/ParameterEvent
Publishers:
  /parameter_events: rcl_interfaces/msg/ParameterEvent
  /rosout: rcl_interfaces/msg/Log
  /turtle1/cmd_vel: geometry_msgs/msg/Twist
Service Servers:
  /teleop_turtle/describe_parameters: rcl_interfaces/srv/DescribeParameters
  /teleop_turtle/get_parameter_types: rcl_interfaces/srv/GetParameterTypes
  /teleop_turtle/get_parameters: rcl_interfaces/srv/GetParameters
  /teleop_turtle/get_type_description: type_description_interfaces/srv/GetTypeDescription
  /teleop_turtle/list_parameters: rcl_interfaces/srv/ListParameters
  /teleop_turtle/set_parameters: rcl_interfaces/srv/SetParameters
  /teleop_turtle/set_parameters_atomically: rcl_interfaces/srv/SetParametersAtomically
Service Clients:

Action Servers:

Action Clients:
  /turtle1/rotate_absolute: turtlesim/action/RotateAbsolute
```

```
somber@LAPTOP-QFH7FQI7: ~  
[+] [v]  
  
Action Servers:  
  /turtle1/rotate_absolute: turtlesim/action/RotateAbsolute  
Action Clients:
```



The screenshot shows the RQT GUI with the RosGraph plugin open. The graph displays the following nodes and their connections:

- /turtle1** (parent container)
 - /turtle1/rotate_absolute** (action client)
 - /turtle1/rotate_absolute_actionfeedback** (action server)
 - /turtle1/cmd_vel_publisher** (publisher)
 - /turtle1/cmd_vel_subscriber** (subscriber)
- /teleop_turtle** (parent container)
 - /teleop_turtle/turtle1** (client)
 - /teleop_turtle/turtle1_velocity** (publisher)
 - /teleop_turtle/turtle1_position** (subscriber)

Connections shown in the graph:

- /turtle1/rotate_absolute** → **/turtle1/rotate_absolute_actionfeedback**
- /turtle1/cmd_vel_publisher** → **/turtle1/cmd_vel_subscriber**
- /teleop_turtle/turtle1** → **/teleop_turtle/turtle1_position**
- /teleop_turtle/turtle1_velocity** → **/teleop_turtle/turtle1**

```
somber@LAPTOP-QFH7FQI7:~/ros2_tasks/task2$ rqt_graph
StandardPaths: wrong permissions on runtime directory /mnt/wslg/runtime-dir, 0777 instead of 0700
```

Related Questions & Answers

Q5. For moving and tracking, what does /teleop_turtle send and receive? Topic names?

A: Publishes velocity commands to `/turtle1/cmd_vel` (`geometry_msgs/msg/Twist`). It does not need to subscribe to pose for basic keyboard teleop.

Q9. Do the arrows in `rqt_graph` match the publisher/subscriber info?

A: Yes. `/teleop_turtle` publishes `/turtle1/cmd_vel`, and `/turtlesim` subscribes to it; `/turtlesim` publishes `/turtle1/pose`.

Task 2 - ROS2 (turtlesim) | Screenshot Set 4

```
somber@LAPTOP-QFH7FQI7:~/ros2_tasks/task2$ ros2 topic list
ros2 topic list -t
/parameter_events
/rosout
/turtle1/cmd_vel
/turtle1/color_sensor
/turtle1/pose
/parameter_events [rcl_interfaces/msg/ParameterEvent]
/rosout [rcl_interfaces/msg/Log]
/turtle1/cmd_vel [geometry_msgs/msg/Twist]
/turtle1/color_sensor [turtlesim/msg/Color]
/turtle1/pose [turtlesim/msg/Pose]
somber@LAPTOP-QFH7FQI7:~/ros2_tasks/task2$ ros2 topic info /turtle1/cmd_vel
Type: geometry_msgs/msg/Twist
Publisher count: 1
Subscription count: 1
somber@LAPTOP-QFH7FQI7:~/ros2_tasks/task2$ ros2 interface show geometry_msgs/msg/Twist
# This expresses velocity in free space broken into its linear and angular parts.

Vector3 linear
float64 x
float64 y
float64 z
Vector3 angular
float64 x
float64 y
float64 z
```

[illegible]

Related Questions & Answers

Q11. Difference between `rostopic list` and `rostopic list -t`?

A: `ros2 topic list` shows only topic names; `-t` shows topic names plus their message types.

Q13. What does `rostopic info /turtle1/cmd_vel`` tell us?

A: It reports the topic type and the number of publishers/subscribers (and related connection info).

Q17. What happened when running ``ros2 topic echo /turtle1/cmd_vel``?

A: It prints nothing until a publisher sends messages; once teleop publishes, Twist messages appear.

Q19. What happened in the echo terminal after pressing an arrow key? Expected?

A: Twist messages appeared/updated (typically linear.x and angular.z change).
Yes, this matches the Twist interface fields.

Q26. What do x, y, z refer to in Twist?

A: They are the components of the linear and angular velocity vectors (linear.{x,y,z} and angular.{x,y,z}). In turtlesim, motion mainly uses linear.x and angular.z.

Task 2 - ROS2 (turtlesim) | Screenshot Set 5



The screenshot displays the ROS2 turtlesim environment. On the left, a window titled 'turtlesim' shows a 2D plot with a blue background and a green turtle. The turtle is moving along a path defined by several lines. On the right, a terminal window shows the output of the 'ros2 topic hz /turtle1/pose' command, which reports an average rate of 62.503 Hz. Below this, the terminal shows the output of the 'ros2 service list' command, listing various services available in the turtlesim environment.

```
somber@LAPTOP-QFH7FQI7:~/ros2_tasks/task2$ ros2 topic hz /turtle1/pose
average rate: 62.503
min: 0.015s max: 0.017s std dev: 0.00055s window: 64
average rate: 62.507
min: 0.015s max: 0.018s std dev: 0.00061s window: 127
average rate: 62.510
min: 0.015s max: 0.018s std dev: 0.00059s window: 190
average rate: 62.513

somber@LAPTOP-QFH7FQI7:~/ros2_tasks/task2$ ros2 topic pub --once /turtle1/cmd_vel geometry_msgs/msg/Twist "{linear: {x: 2.0, y: 0.0, z: 0.0}, angular: {x: 0.0, y: 0.0, z: 1.8}}"
publisher: beginning loop
publishing #1: geometry_msgs.msg.Twist(linear=geometry_msgs.msg.Vector3(x=2.0, y=0.0, z=0.0), angular=geometry_msgs.msg.Vector3(x=0.0, y=0.0, z=1.8))

somber@LAPTOP-QFH7FQI7:~/ros2_tasks/task2$ ros2 service list
/clear
/kill
/reset
/spawn
/teleop_turtle/describe_parameters
/teleop_turtle/get_parameter_types
/teleop_turtle/get_parameters
/teleop_turtle/get_type_description
/teleop_turtle/list_parameters
/teleop_turtle/set_parameters
/teleop_turtle/set_parameters_atomically
/turtle1/set_pen
/turtle1/teleport_absolute
/turtle1/teleport_relative
/turtlesim/describe_parameters
/turtlesim/get_parameter_types
/turtlesim/get_parameters
/turtlesim/get_type_description
/turtlesim/list_parameters
/turtlesim/set_parameters
/turtlesim/set_parameters_atomically
somber@LAPTOP-QFH7FQI7:~/ros2_tasks/task2$
```

Related Questions & Answers

Q22. At what rate is the simulator publishing /turtle1/pose?

A: About 62.5 Hz (measured by `ros2 topic hz /turtle1/pose`).

Q26. What do x, y, z refer to in the published cmd_vel Twist?

A: They are linear and angular velocity components; in 2D turtlesim, linear.x sets forward speed and angular.z sets turn rate (yaw).

Task 2 - ROS2 (turtlesim) | Screenshot Set 6

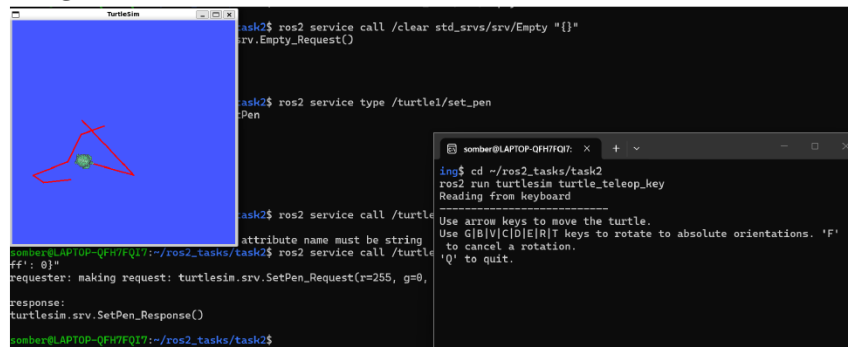
```
somber@LAPTOP-QFH7FQI7:~/ros2_tasks/task2$ ros2 service list -t
/clear [std_srvs/srv/Empty]
/kill [turtlesim/srv/Kill]
/reset [std_srvs/srv/Empty]
/spawn [turtlesim/srv/Spawn]
/teleop_turtle/describe_parameters [rcl_interfaces/srv/DescribeParameters]
/teleop_turtle/get_parameter_types [rcl_interfaces/srv/GetParameterTypes]
/teleop_turtle/get_parameters [rcl_interfaces/srv/GetParameters]
/teleop_turtle/get_type_description [type_description_interfaces/srv/GetTypeDescription]
/teleop_turtle/list_parameters [rcl_interfaces/srv/ListParameters]
/teleop_turtle/set_parameters [rcl_interfaces/srv/SetParameters]
/teleop_turtle/set_parameters_atomically [rcl_interfaces/srv/SetParametersAtomically]
/turtle1/set_pen [turtlesim/srv/SetPen]
/turtle1/teleport_absolute [turtlesim/srv/TeleportAbsolute]
/turtle1/teleport_relative [turtlesim/srv/TeleportRelative]
/turtlesim/describe_parameters [rcl_interfaces/srv/DescribeParameters]
/turtlesim/get_parameter_types [rcl_interfaces/srv/GetParameterTypes]
/turtlesim/get_parameters [rcl_interfaces/srv/GetParameters]
/turtlesim/get_type_description [type_description_interfaces/srv/GetTypeDescription]
/turtlesim/list_parameters [rcl_interfaces/srv/ListParameters]
/turtlesim/set_parameters [rcl_interfaces/srv/SetParameters]
/turtlesim/set_parameters_atomically [rcl_interfaces/srv/SetParametersAtomically]
somber@LAPTOP-QFH7FQI7:~/ros2_tasks/task2$
```

```
somber@LAPTOP-QFH7FQI7:~/ros2_tasks/task2$ ros2 interface show std_srvs/srv/Empty
---
somber@LAPTOP-QFH7FQI7:~/ros2_tasks/task2$ ros2 service call /clear std_srvs/srv/Empty "{}"
requester: making request: std_srvs.srv.Empty_Request()
```

```
response:
std_srvs.srv.Empty_Response()

somber@LAPTOP-QFH7FQI7:~/ros2_tasks/task2$ ros2 service type /turtle1/set_pen
ros2 interface show turtlesim/srv/SetPen
turtlesim/srv/SetPen
uint8 r
uint8 g
uint8 b
uint8 width
uint8 off
---
```

change color



```
somber@LAPTOP-QFH7FQI7:~/ros2_tasks/task2$ ros2 service call /clear std_srvs/srv/Empty "{}"
requester: making request: std_srvs.srv.Empty_Request()

somber@LAPTOP-QFH7FQI7:~/ros2_tasks/task2$ ros2 service type /turtle1/set_pen
turtlesim/srv/SetPen

somber@LAPTOP-QFH7FQI7:~/ros2_tasks/task2$ ros2 service call /turtle1/set_pen
requester: making request: turtlesim.srv.SetPen_Request(r=255, g=0, b=0)
response:
turtlesim.srv.SetPen_Response()

somber@LAPTOP-QFH7FQI7:~/ros2_tasks/task2$
```

Related Questions & Answers

Q2. How can you see the service type for /clear?

A: Either use `ros2 service list -t` to list services with types, or `ros2 service type /clear` to query the type directly.

Q3. Looking at the interface for the /clear service type, what does it imply?

A: /clear uses std_srvs/srv/Empty, meaning the request has no fields (it just triggers an action).

Q5. What happened when calling /clear?

A: The turtlesim window clears the drawn trails/lines.

Challenge: Change the turtle line color to red.

A: Call /turtle1/set_pen (turtlesim/srv/SetPen) with r=255, g=0, b=0 (and off=0) so new trails are red.

Task 2 - ROS2 (turtlesim) | Screenshot Set 7

```
sambor@LAPTOP-QFH7FQ17:~/ros2_tasks/task2$ ros2 service call /turtle1/set_pen turtlesim/srv/SetPen "{r: 255, g: 0, b: 0, width: 3, 'off': 0}"
requester: making request: turtlesim.srv.SetPen_Request(r=255, g=0, b=0, width=3, off=0)
response:
turtlesim.srv.SetPen_Response()
sambor@LAPTOP-QFH7FQ17:~/ros2_tasks/task2$ ros2 action list -t
/turtle1/rotate_absolute [turtlesim/action/RotateAbsolute]
sambor@LAPTOP-QFH7FQ17:~/ros2_tasks/task2$ ros2 action info /turtle1/rotate_absolute -t
Action: /turtle1/rotate_absolute
Action clients: 1
  /teleop_turtle [turtlesim/action/RotateAbsolute]
Action servers: 1
  /turtlesim [turtlesim/action/RotateAbsolute]
sambor@LAPTOP-QFH7FQ17:~/ros2_tasks/task2$ ros2 action send_goal /turtle1/rotate_absolute turtlesim/action/RotateAbsolute "{theta: 1.57}"
Waiting for an action server to become available...
Sending goal:
  theta: 1.57
Goal accepted with ID: 0d9ba9960e69422ba8a9574856753f
Result:
  delta: -1.8240000009536743
Goal finished with status: SUCCEEDED
sambor@LAPTOP-QFH7FQ17:~/ros2_tasks/task2$ |
sambor@LAPTOP-QFH7FQ17:~/ros2_tasks/task2$ ros2 action send_goal /turtle1/rotate_absolute turtlesim/action/RotateAbsolute "{theta: 1.57}" --feedback
Waiting for an action server to become available...
Sending goal:
  theta: 1.57
Feedback:
  remaining: 0.014858841896857129
Goal accepted with ID: 4eeb9c8353f8408faf169a36dd49b0c3
Result:
  delta: 0.0
Goal finished with status: SUCCEEDED
```

Related Questions & Answers

Q3. Which node is the server and which is the client for the rotate action?

A: For /turtle1/rotate_absolute, /turtlesim is the action server; /teleop_turtle is listed as an action client (and the `ros2 action send\_goal` command also acts as a client while it runs).